

IBM University Engagement Project Submission

“Exploring the Power of Agentic AI with IBM Llama ”

Project Tile – [AgroGranie AI-Powered Smart Farming Advisor]

Domain of Project – [Agriculture]

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Problem Statement -

Problem Statement: AI Agent for Smart Farming Advice

The Challenge

An AI Agent for Smart Farming Advice, powered by RAG (Retrieval-Augmented Generation), supports small-scale farmers by delivering real-time, localized agricultural guidance. It retrieves trusted data on weather forecasts, soil conditions, crop recommendations, pest control measures, and current market prices from agricultural departments, meteorological sources, and agri-tech platforms.

The Objective

- To develop an **Agentic AI-powered Smart Farming Assistant** using **IBM watsonx.ai, IBM Llama Models, and IBM Cloud** to support farmers with intelligent agricultural guidance.
- To provide **real-time, personalized recommendations** for crop selection, weather forecasting, disease management, government schemes, and mandi prices through specialized AI agents.
- To integrate **Retrieval-Augmented Generation (RAG)** with trusted agricultural knowledge sources, ensuring accurate and reliable responses for farmers.
- To support **multilingual text and voice interaction**, enabling farmers to communicate naturally using IBM Speech to Text and IBM Text to Speech services.
- To implement **Farmer Memory and Explainable AI**, delivering personalized, transparent, and evidence-based recommendations for better decision-making.
- To build a **secure, scalable, and cloud-native solution** on IBM Cloud that improves farming productivity, reduces risk, and supports sustainable agriculture.

Proposed Solution – AgroGenie AI: Agentic Smart Farming Advisor

- The proposed solution is an AI-powered **Smart Farming Assistant** developed using **IBM watsonx.ai, IBM Granite Models, IBM Cloud, Retrieval-Augmented Generation (RAG), IBM Cloud Object Storage, and Speech Services**. The system is designed to assist farmers in making better decisions by providing personalized recommendations related to crop selection, weather conditions, plant diseases, government schemes, and market prices through a single intelligent platform.
- The solution follows a **multi-agent architecture**, where each AI agent performs a specific task while working together to provide accurate and context-aware responses. The **Crop Recommendation Agent** suggests suitable crops based on soil type, weather conditions, irrigation facilities, and farmer preferences. The **Weather Intelligence Agent** provides real-time weather forecasts along with farming advisories. The **Disease Advisory Agent** analyzes uploaded crop images, combines weather conditions and agricultural knowledge, and recommends suitable treatment and preventive measures. The **Mandi Price Agent** provides current market prices to help farmers make better selling decisions, while the **Government Scheme Agent** recommends relevant schemes based on the farmer's profile.
- To improve accuracy, the system uses a **Retrieval-Augmented Generation (RAG)** pipeline that retrieves verified agricultural information before generating responses. A **Farmer Memory Module** stores user preferences and previous interactions to provide personalized recommendations over time. An **Explainability Engine** displays the reason behind every recommendation, the supporting evidence, and the confidence level, making the system more transparent and trustworthy.
- The application also supports **multilingual text, voice input, and voice output** using IBM Speech to Text and Text to Speech services, making it accessible for farmers with different language preferences. The complete solution is deployed using **IBM Cloud** services, providing a secure, scalable, and production-ready platform for intelligent agriculture.

Technology Used

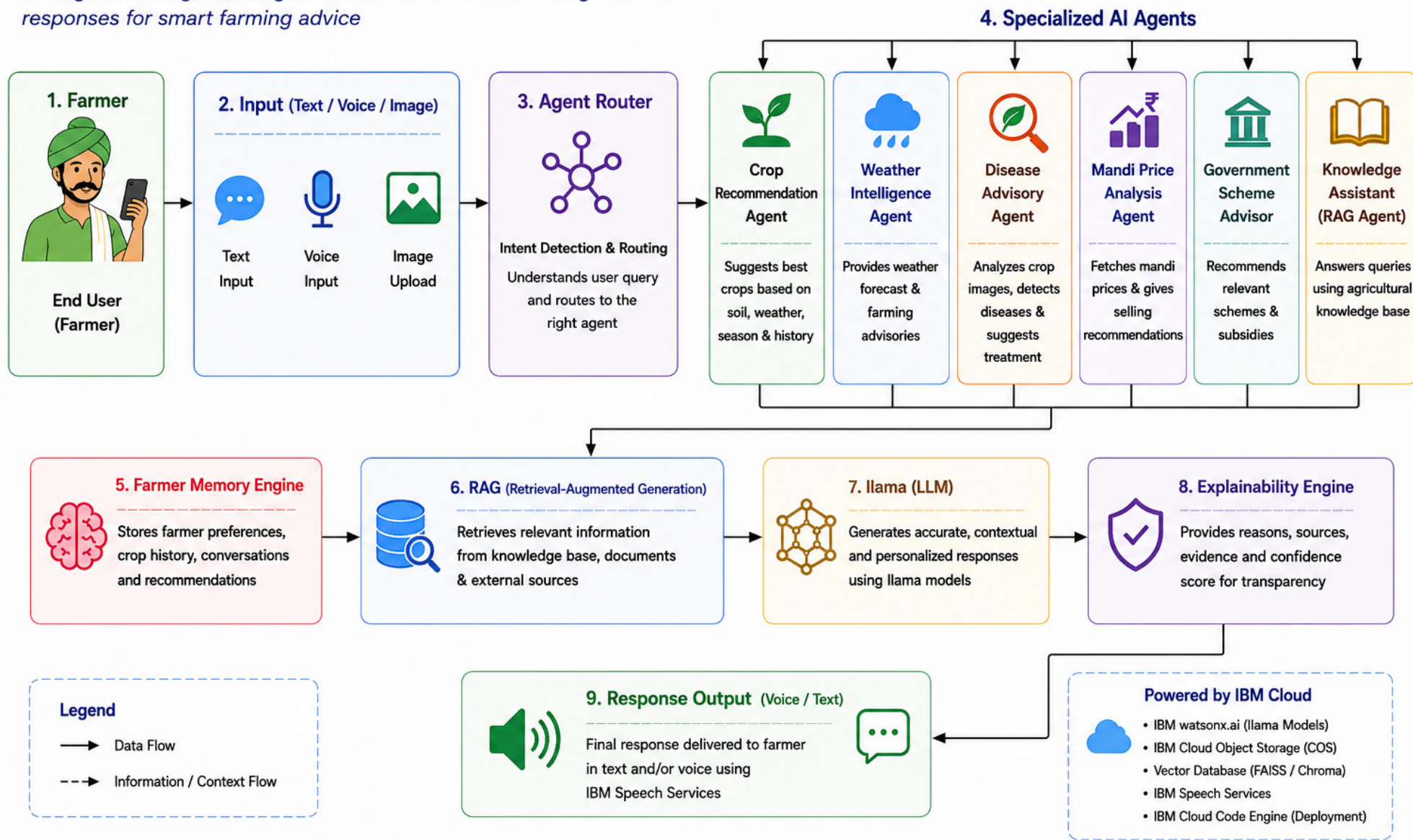
- **IBM watsonx.ai** – Used for deploying and managing AI models, prompt orchestration, and integrating IBM Granite models for intelligent agricultural assistance
- **IBM Llama Model (llama-3-3-70b-instruct)** – Used for natural language understanding, reasoning, multilingual conversations, explainability, and personalized farming recommendations.
- **IBM Cloud** – Provides the cloud infrastructure to securely build, deploy, monitor, and scale AgroGenie AI using IBM Code Engine and managed cloud services.
- **IBM Cloud Object Storage (COS)** – Stores uploaded crop images securely for disease analysis and manages agricultural documents and resources.
- **IBM Speech to Text** – Converts farmers' voice queries into text, enabling multilingual voice-based interaction with the assistant.
- **IBM Text to Speech** – Converts AI-generated responses into natural speech, improving accessibility for farmers.
- **Retrieval-Augmented Generation (RAG)** – Retrieves verified agricultural knowledge before generating responses, ensuring accurate and reliable recommendations.
- **MongoDB Atlas** – Stores farmer profiles, conversation history, memory, recommendations, and application data for personalized interactions.
- **Docker & IBM Code Engine** – Used for containerization and deployment of AgroGenie AI on IBM Cloud, ensuring scalability and production readiness.

AgroGenie AI – Multi-Agent Workflow

- AgroGenie AI follows a custom **Agentic AI workflow** where every user request is processed through an intelligent routing mechanism. The system first accepts user input in the form of **text, voice, or crop images**. The **Agent Router** analyzes the user's intent and forwards the request to the most appropriate specialized AI agent.
- Each specialized agent performs a dedicated task such as **Crop Recommendation, Weather Intelligence, Disease Advisory, Mandi Price Analysis, Government Scheme Assistance, or Knowledge Retrieval**. These agents interact with IBM Granite models, Retrieval-Augmented Generation (RAG), Farmer Memory, and external APIs to generate accurate and personalized responses.
- Before generating the final response, the Explainability Engine combines evidence from multiple agents and presents transparent recommendations with confidence scores and supporting information. Finally, the response is delivered to the farmer in text or voice using IBM Speech Services.

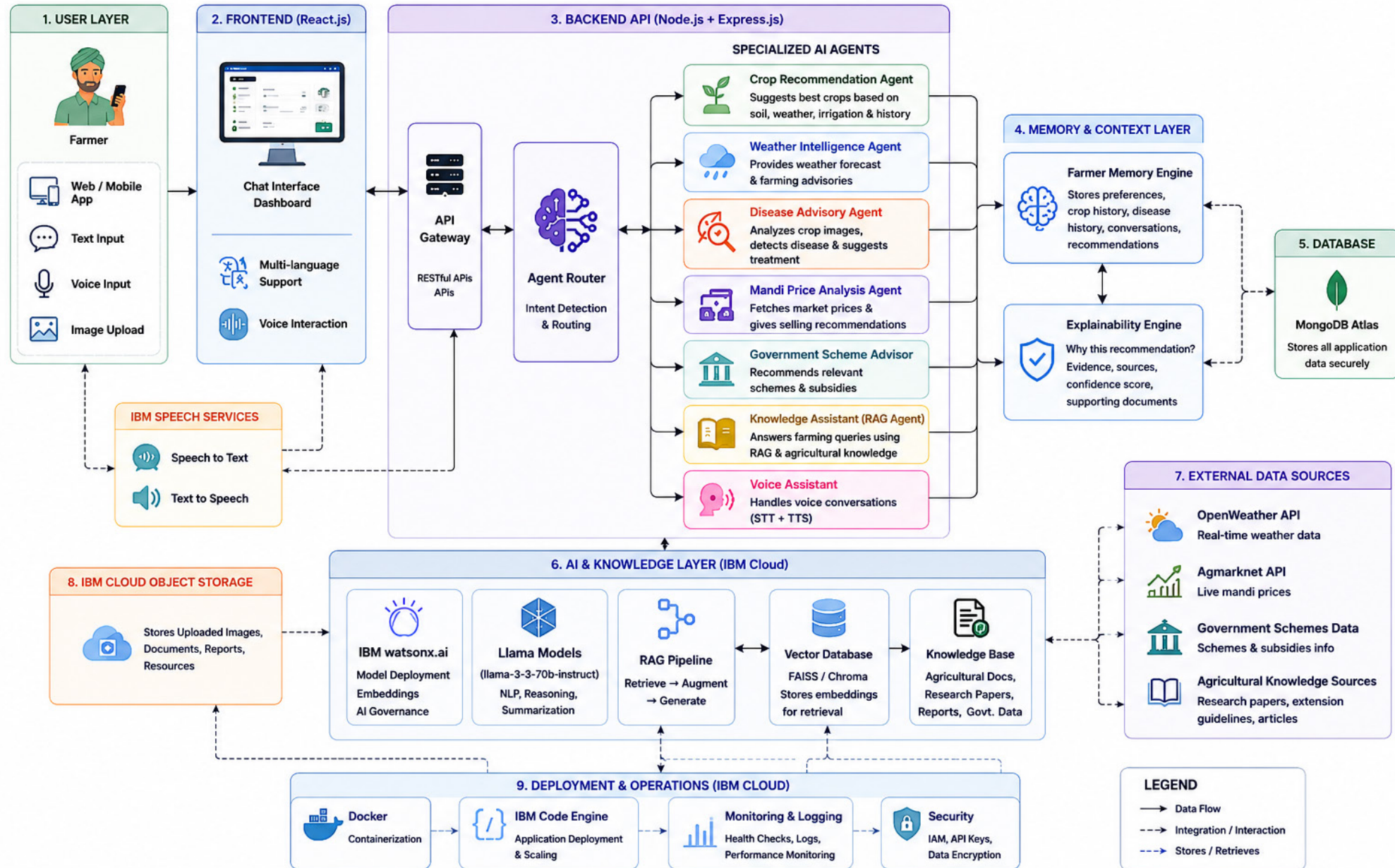
AgroGenie AI – Multi-Agent Workflow

Intelligent routing, multi-agent collaboration and knowledge-driven responses for smart farming advice



AgroGenie AI – Agentic Smart Farming Advisor

Architecture Diagram



Role of Agentic AI in the solution

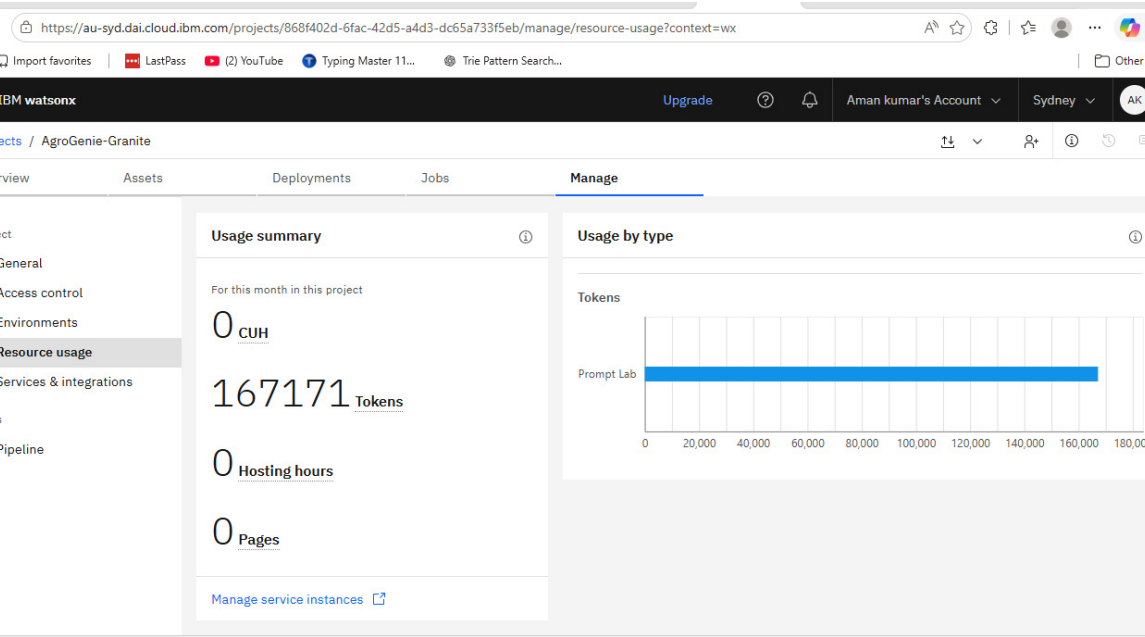
- **Intelligent Task Routing:** The **Agent Router** analyzes the farmer's query and automatically forwards it to the most suitable specialized AI agent.
- **Specialized Multi-Agent Collaboration:** Independent agents for **Crop Recommendation, Weather Intelligence, Disease Advisory, Mandi Price Analysis, Government Schemes, Knowledge Assistant, and Voice Assistant** work together to solve agricultural problems.
- **Context-Aware Decision Making:** Each agent uses **Farmer Memory**, previous interactions, weather conditions, crop history, and user preferences to provide personalized recommendations.
- **Knowledge-Augmented Responses:** The **Knowledge Agent** integrates **Retrieval-Augmented Generation (RAG)** with IBM Llama to retrieve verified agricultural information before generating responses.
- **Collaborative Reasoning:** Multiple agents combine their outputs, such as weather forecasts, disease analysis, market prices, and government schemes, to provide comprehensive farming advice instead of isolated responses.
- **Explainable AI:** The **Explainability Engine** clearly shows the reasoning, supporting evidence, confidence score, and contributing agents behind every recommendation.
- **Scalable Agentic Architecture:** The modular design allows new agricultural agents and IBM Cloud services to be integrated easily without affecting existing functionalities.



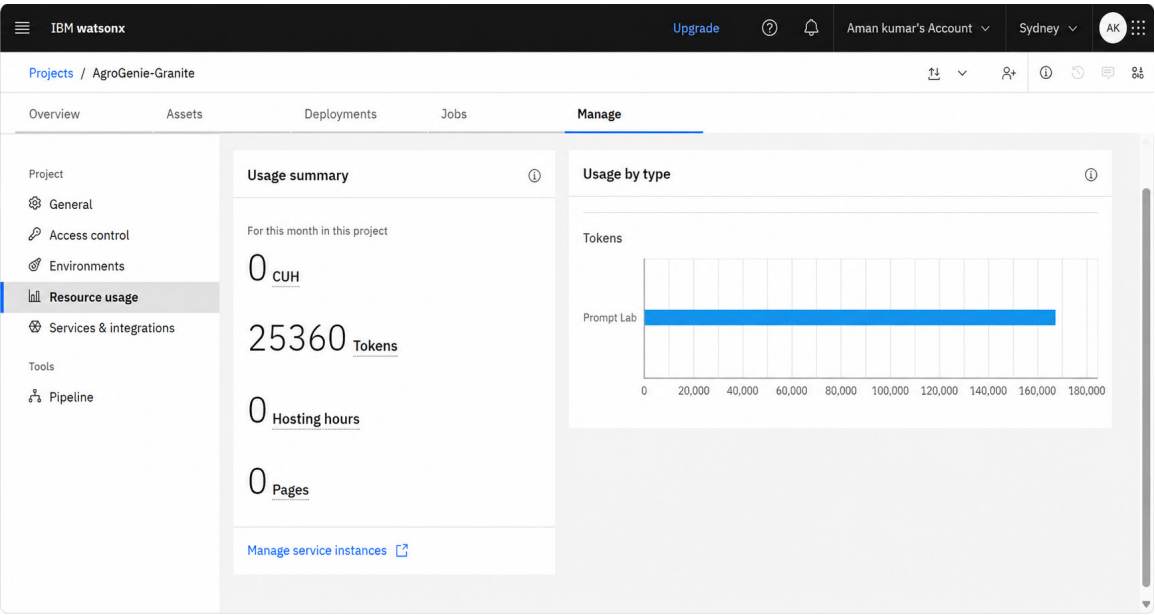
Token Usage

141,811 token are used

Before Prompt



After the Prompt



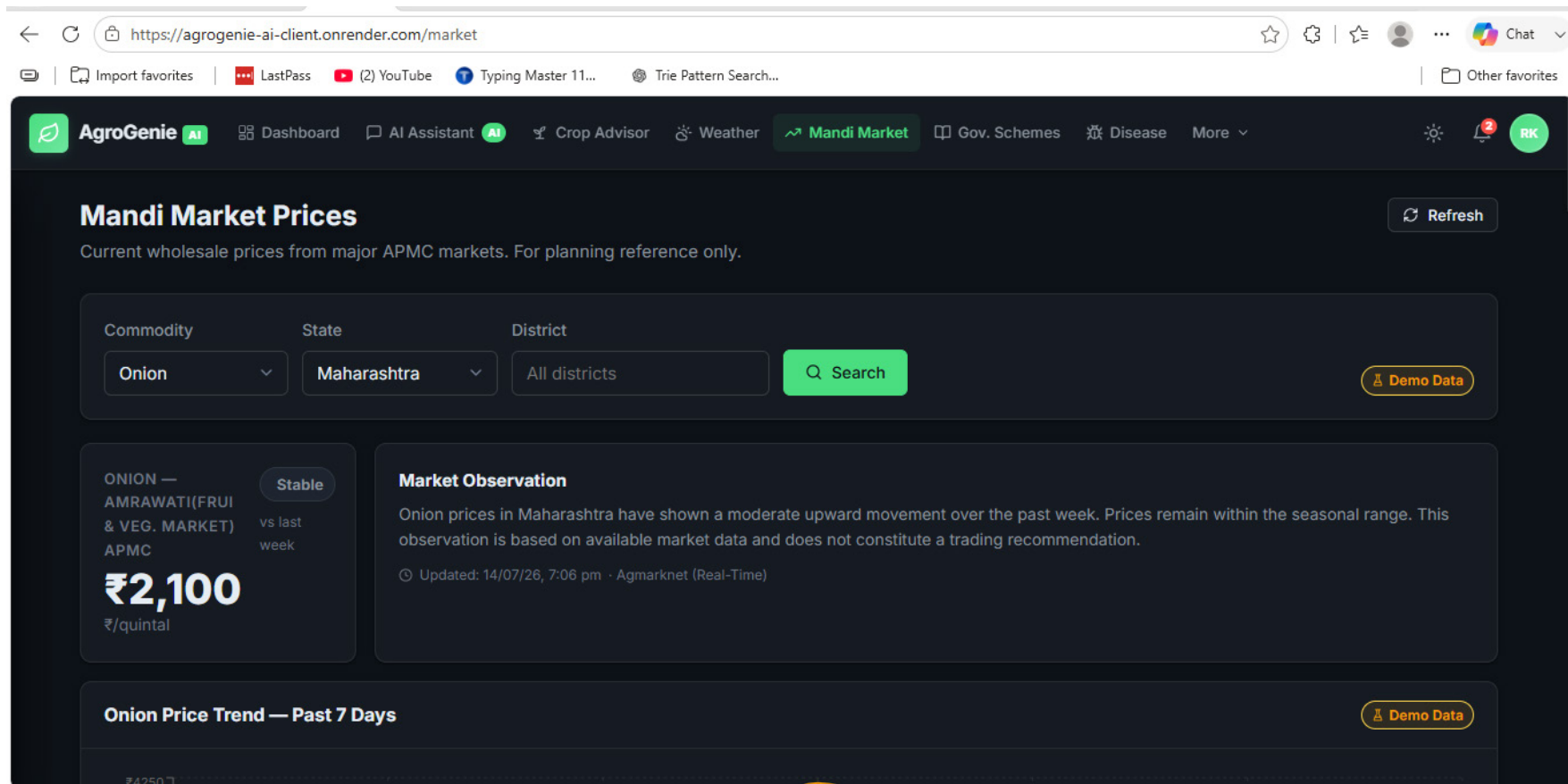
Novelty and Uniqueness

- Developed a **single AI platform** that combines crop recommendation, disease detection, weather forecasting, mandi price analysis, government schemes, and agricultural knowledge instead of using separate applications.
- Implemented a **Multi-Agent AI Architecture**, where specialized agents collaborate to solve farming problems, making the system more accurate and efficient.
- Integrated **Farmer Memory** to remember user preferences, crop history, and previous interactions, enabling personalized recommendations in future conversations.
- Built an **Explainability Engine** that provides the reason, supporting evidence, and confidence level behind every recommendation, improving transparency and user trust.
- Combined **RAG (Retrieval-Augmented Generation)** with IBM Llama Models to retrieve verified agricultural information before generating responses, reducing incorrect or misleading answers.
- Supports **multilingual text, voice input, and voice output** using IBM Speech Services, making the solution more accessible for farmers with different language preferences.
- Designed as a **cloud-native solution** using IBM Cloud services, making it secure, scalable, and easy to deploy for real-world agricultural applications.

Git Hub Link

Live Demo of Project - <https://agrogenie-ai-client.onrender.com>

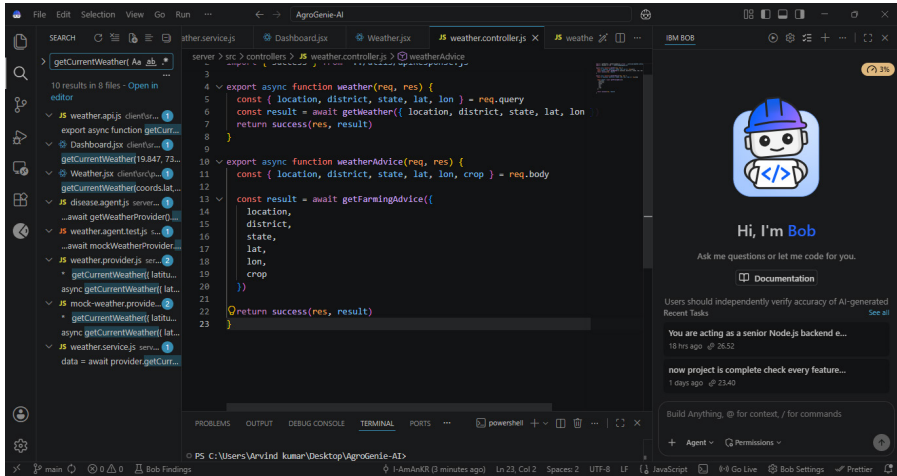
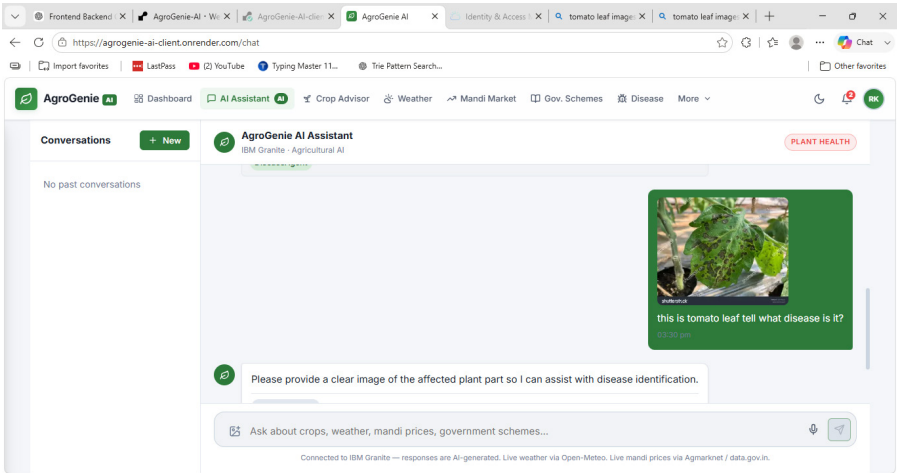
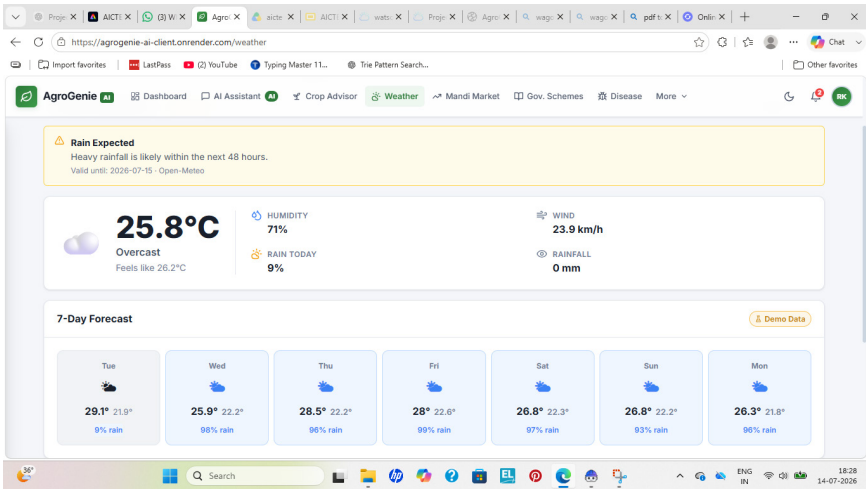
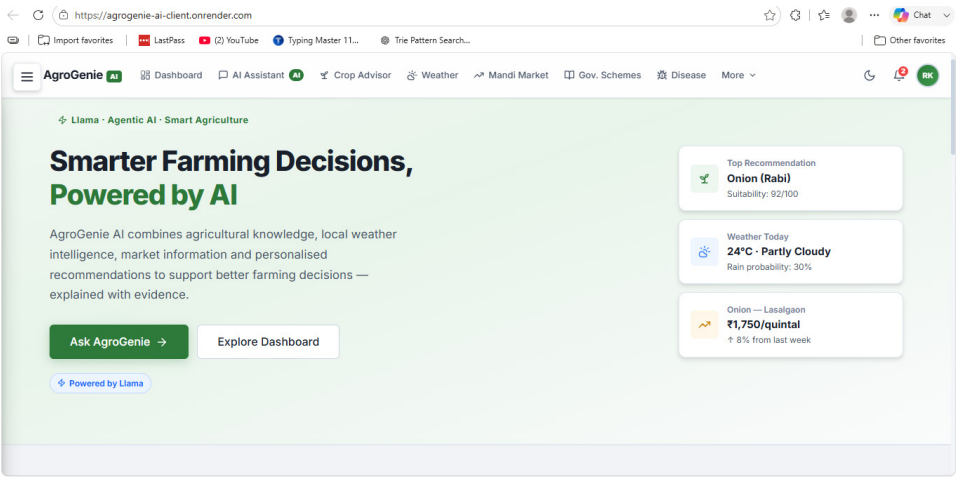
Github URL – <https://github.com/I-AmAnKR/AgroGenie-AI>



Future Scope

- Develop a **mobile application** to provide farmers with easier access to AgroGenie AI in remote areas.
- Integrate **IoT sensors and smart farming devices** for real-time monitoring of soil moisture, temperature, humidity, and crop health.
- Add **satellite imagery and drone-based crop monitoring** for early disease detection and precision farming.
- Expand support for **more regional Indian languages and dialects** to improve accessibility for farmers across different states.
- Introduce **predictive analytics** for crop yield estimation, pest outbreaks, and long-term weather forecasting using historical data.
- Integrate with **additional government portals and agricultural databases** to provide more schemes, subsidies, and advisory services.
- Build a **farmer community platform** where users can share experiences, best practices, and receive expert guidance.
- Deploy AgroGenie AI at **state and national levels** through collaboration with agricultural departments, Krishi Vigyan Kendras (KVKs), and Farmer Producer Organizations (FPOs).

Project Demo Images



Certificates Earned:

In recognition of the commitment to achieve
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Has successfully satisfied the requirements for:

Getting Started with Artificial Intelligence



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Learning hours: 30 mins

Thank You!

Thank you for your time and interest.